RIT

Piter Garcia <pzg8794@g.rit.edu>

Research Topic Proposal for IDAI-700

Piter Garcia (RIT Student) <pzg8794@rit.edu> Fri, Sep 12, 2025 at 2:02 AM
Reply-To: pzg8794@rit.edu
To: Evan Selinger <emsgsh@rit.edu>

Hi Professor Selinger,

Thank you for this incredibly helpful response!

You've clarified several important aspects that will strengthen my approach.

I completely understand your point about "equity" being one interpretation of fairness rather than synonymous with it. And, you're absolutely right that I'll need to philosophically justify why I see equity as foundational—encompassing transparency, accountability, and ultimately leading to trust. This gives me a clear direction for the theoretical framework of my paper.

Regarding literature breadth, I think my previous point didn't come across clearly. When I mentioned "many papers, tests, and analyses," I was referring specifically to the empirical results and data analyses from studies in data science, bioinformatics, and healthcare fields. I plan to use specific test results and statistical analyses from these papers—not the entire papers—to support and validate my argument.

In data-heavy fields like healthcare informatics, drawing on multiple studies' findings is standard practice for building a robust evidence base. My goal is to use these empirical results to give the terminology I employ (like "equity") a more concrete, measurable context, helping distinguish my findings and prevent misinterpretation of key concepts.

I appreciate your patience and guidance throughout this process. You've helped me refine my research question and approach to making a focused, compelling argument within the scope of this class.

Looking forward to developing this further!

Best,
Piter

On Wed, Sep 10, 2025 at 10:50 AM Evan Selinger <emsgsh@rit.edu> wrote:

> Hi Piter,
>
> Here are my top-of-mind thoughts.
>
> 1. I now have a better sense of why you don’t see the distinction between chronic and acute illnesses as relevant here. You are 100% right that all patients deserve fair treatment.
>
> 2. One thing to think about is that “equity” is but one of many approaches to “fairness.” It shouldn’t be thought of as synonymous with how everyone understands fairness. For example, equity can mean providing different levels of support to achieve equal outcomes, such as allocating more resources to those who need them the most or who have been historically disadvantaged. And this approach can conflict with someone who understands fairness as treating everyone the same, regardless of their circumstances. As an example, think about the current debate over whether colleges should be able to offer affirmative action considerations for admissions.
>
> My point is simply that ethical ideals like “equity” and “fairness” can be interpreted in different, sometimes conflicting ways. In fact, these issues also apply to many other contested moral concepts, including “freedom,” “justice,” and “rights,” among others.
>
> Bottom line: For purposes of your paper, you are free to use what you see as the most appropriate understanding of “equity.” I’m simply drawing your attention to the complexity of the term.

3. Given #2, it's quite possible that "equity" is your foundational ethical concept—that it encompasses the others that you deem relevant. If so, part of your task is to philosophically justify this argument. In other words, if you see some ethical principles as less fundamental than others (or being encompassed by others), you'll need to make a case for why. This is one way to do excellent philosophical work!

4. As to the breadth of the literature, please keep one thing in mind. Yes, ultimately, addressing concerns like equity in healthcare requires a dialogue among disciplines and extends to a range of actors and institutions! But a good research paper for this class won't try to solve everything. You need a book (at least!) for that. So, the more you try to address, the more likely it is that your analysis will tilt away from making arguments that are compelling and detailed.

Does this make sense?

Best,
Evan

Evan Selinger
Professor, Philosophy Dept.
Rochester Institute of Technology
Office: Eastman (Building 1), Room 3146
(585) 475-2531 http://eselinger.org

From: Piter Garcia (RIT Student) <pzg8794@rit.edu>
Sent: Wednesday, September 10, 2025 12:57 AM
To: Evan Selinger <emsgsh@rit.edu>
Subject: Re: Research Topic Proposal for IDAI-700

Hi Professor Selinger,

Thank you for your thoughtful feedback!

Let's clarify my research focus and address the points you've raised in your email.

1 - Research Scope and Personal Experience Context
I shared my personal experience not to define the paper's scope, but to highlight **why fairness in healthcare AI is essential**. My journey illustrates a systemic problem: when patients—particularly those from CLD communities—aren't heard or believed, healthcare becomes ineffective regardless of the specific diagnosis. *This is fundamentally about fairness and equity, not medical categorization.*

The type of chronic illness isn't the focus because fairness principles apply universally. Whether someone has diabetes, fibromyalgia, or another condition, they deserve transparent, accountable care. My experience—*being dismissed as "lazy" or having symptoms attributed solely to depression*—**reflects broader systemic biases that AI could either reinforce or help dismantle.**

2 - Ethical Framework Selection
You raise an excellent point about my emphasis on **transparency, accountability, and trust**. Upon reflection, I believe **equity** may serve better as the foundational principle, as it inherently encompasses fairness and inclusivity. This would refine my research question to:

"How can we design equitable AI systems that healthcare providers and patients from culturally and linguistically diverse (CLD) communities can trust?"

This framing positions equity as the overarching goal, with transparency and accountability as mechanisms for achieving trustworthy, inclusive systems.

3 - Literature Breadth
The interdisciplinary nature of this work, *spanning AI/technology, healthcare, and human development*, naturally requires diverse sources. In my experience working in Data Science and Bioinformatics, this level of breadth is not only common but necessary—especially now, as both fields are booming and reshaping how we understand equity in healthcare. However, I'm focused on synthesizing high-quality sources that offer strong analyses and relevant findings, rather than limiting scope for the sake of uniformity.

If any of this still feels unclear, I'd be happy to meet via Zoom. Sometimes email isn't the best medium for this kind of exchange, and I want to make sure my perspective—and the care behind it—comes through.

Thank you again for your guidance and willingness to engage so thoughtfully.

Best,
Piter

On Tue, Sep 9, 2025 at 6:41 PM Evan Selinger <emsgsh@rit.edu> wrote:

> Hi Piter,
>
> Thanks for the ongoing discussion!
>
> As always, I appreciate that you're trying to integrate your personal experience of systemic exclusion from the healthcare system with your interest in designing transparent and accountable AI systems that healthcare providers and patients from culturally and linguistically diverse communities can trust.
>
> However, there are a few things in your response that I have questions about.
>
> 1. Since you have had experience with a chronic illness and this experience is motivating your research paper, I think it's important to acknowledge a few things. Chronic illness can have different characteristics from acute ones. And the differences among chronic illnesses can be profound. In short, details and specificity can make a considerable difference when it comes to receiving appropriate healthcare. And yet, your research question is quite broad; it focuses on the so-called "healthcare system" in general. This difference makes me wonder if your research question is sufficiently scoped.
>
> 2. Your proposal focused on three ethical terms: "transparency," "accountability," and "trust." This selection begs the question of why you are focusing on these three ethical principles, as opposed to others. Do you see them as fundamental? Is there a specific reason that you are focusing on these three? One reason I'm asking is that the titles of the papers you list include additional ethical principles—"equity," "inclusivity," and "fairness." In other words, the papers suggest a broader inquiry than your initial research question.
>
> 3. Your papers, based on their titles, cover a vast range of material—everything from community engagement to clinician responsibility and regulatory consequences. I'm not sure what the common ground is, and I wonder if you're potentially considering sources that are too varied from one another to form a coherent and targeted background for a meaningful research paper.
>
> Does this make sense?
>
> Best,
> Evan
>
> **Evan Selinger**
> Professor, Philosophy Dept.
> Rochester Institute of Technology
> Office: Eastman (Building 1), Room 3146

(585) 475-2531 http://eselinger.org

From: Piter Garcia (RIT Student) <pzg8794@rit.edu>
Sent: Tuesday, September 9, 2025 4:38 PM
To: Evan Selinger <emsgsh@rit.edu>
Subject: Re: Research Topic Proposal for IDAI-700

Hi Professor Selinger,

I appreciate your thoughtful guidance!

I have already begun exploring the literature and see encouraging signals that my refined thesis can be supported.

My Research Foundation

First, let me shed light on what I already know exists. In my RIT research portfolio, I have engaged with fairness-aware computational biology frameworks that explicitly integrate transparency and accountability in diagnostic algorithms. I felt comfortable doing that because of three things: 1) I was the subject—in my case, healthcare was the problem (what caused my chronic illness), but as a scientist, I rationalized and understood that healthcare was not the problem but the means by which it was administered; 2) I was the solution—in my case, healthcare did not have any solution (therefore chronic), but as a scientist, I realized that it was my mindset, research, and willpower that would give my body the chance to recover; 3) I had the exposure—luckily, I was able to work in the bio and health informatics area of my field (CS, DS, and AI), which led me to work and collaborate with other scientists who were actively working on some sort of fairness approach to make healthcare more accessible and promoting the idea of preventive care.

Educational Equity Insights

Second, at some point during my journey at RIT, I started to understand and merge important aspects of my health via my work at the University of Rochester, where I engaged in educational equity research, emphasizing the importance of trust in systems serving culturally and linguistically diverse (CLD) families. I felt comfortable doing that because, for the first time in my life, I understood how rigid the educational system was. Here you have a person who teaches the subject he is taking, yet he couldn't even read well the first five questions, let alone answer the ten questions, due to an assessment style that is far from inclusive, which led to several symptoms that I was later able to identify and treat accordingly.

This was what led me to identify the vicious cycle I was living in: deception leading to stress, stress aggravating my chronic illness, and depression drawing me into what I could be. However, throughout this journey, I found that I am not alone, and that kids, especially from CLD families, have it a lot worse because of the lack of resources. The research on this alone is more than sufficient, and I believe it surpasses the basis for linking transparency, accountability, and trust to equity outcomes.

Research Viability and Personal Responsibility

Lastly, as far as my preliminary review goes—given the topic and the goal—in my opinion, the scholarly community has done enough background work to make the question not just viable but a necessity. Weighing my background against the field (AI) and the subject (CLD individuals like me), I feel a responsibility to take on this topic and approach it from an angle of fairness. Similarly, given the current climate we are in, I intend to bridge these three important aspects of our lives (fairness, technology, and community) through a lens of fairness—shining a light on critical factors that have always been the root of injustice—guided by experienced reasoning and not fear.

Literature Review Findings

Throughout my research, these are some of the peer-reviewed literature I have come across that I find very relevant to my topic:

Highest Relevance (Directly on CLD + Trust in Healthcare AI):

- Achieving Health Equity through Conversational AI (2024, PMC) – Centers CLD voices, patients, and clinicians in equitable AI design.
- Promoting Cultural Inclusivity in Healthcare AI (2023, HMPI) – Provides a framework for embedding cultural inclusivity into healthcare AI.
- Examining Inclusivity: The Use of AI and Diverse Populations in Health and Social Care (2025, BMC MIDM) – Looks at inclusivity and regulatory concerns for underrepresented groups.
- Health Equity and Ethical Considerations in AI in Public Health (2024, CDC) – Highlights inclusive data, transparency, and community engagement.

Strong Relevance (Transparency, Accountability, Provider Trust):

- Evaluating Accountability, Transparency, and Bias in AI (2025, BMC Medical Ethics) – Discusses clinician responsibility, bias, and transparency in healthcare AI.
- Holding AI to Account: Trustworthy AI in Healthcare (2022, arXiv) – Ethnographic insights on accountability, organizational trust, and clinician relationships.

Supporting Relevance (Broader Ethical/Conceptual Foundations):

- Transparency and Accountability in AI Systems (2024, Frontiers in Human Dynamics) – Lays out broad strategies for transparency and accountability across domains.
- Towards FATE in AI for Social Media and Healthcare (2023, arXiv) – Systematic review of Fairness, Accountability, Transparency, and Ethics (FATE), adaptable to healthcare.

These are only some of the relevant peer-reviewed literature I have come across. However, I may or may not reference them all—feel free to suggest otherwise—because there are more papers I plan to reference due to the extensive research I have done previously on related topics. This way, I can guarantee that my thesis is principled, succinct, and well-grounded in a sufficient body of literature and experience.

I'll keep you updated as I gather and put together my sources. Thank you again for steering me toward this next step, which I will use as part of my reflection journal in this course.

Best regards,
Piter
On Mon, Sep 8, 2025 at 5:37 PM Evan Selinger <emsgsh@rit.edu> wrote:

> Hi Piter,
>
> I like it! The question is whether this version of the thesis, which is principled and succinct, aligns with a sufficient amount of peer-reviewed literature. In other words, you've reached the stage where it's time to look for relevant articles. It's an empirical question whether the scholarly community has done enough background work to make your question viable at this time.
>
> So, here's what I recommend.
>
> 1. Start using RIT's library (and other related sources if you want) to see if this approach works. If so, huzzah!
>
> 2. If not, the process of looking at the material that comes up during the search should offer context clues to help you successfully refine your thesis. And if you have to refine your thesis, you'll repeat Step #1.
>
> Again, the key point to keep in mind is that your revised research question still centers on three ethical terms: transparency, accountability, and trust. Articles that don't give these terms due ethical consideration won't be helpful.
>
> Please let me know how the process goes.
>
> Best,
> Evan
>
> **Evan Selinger**
> Professor, Philosophy Dept.
> Rochester Institute of Technology
> Office: Eastman (Building 1), Room 3146
> (585) 475-2531 http://eselinger.org
>
> ---
>
> **From:** Piter Garcia (RIT Student) <pzg8794@rit.edu>
> **Sent:** Monday, September 8, 2025 5:09 PM
> **To:** Evan Selinger <emsgsh@rit.edu>
> **Subject:** Re: Research Topic Proposal for IDAI-700

Hi Professor Selinger,

Thank you again for your thoughtful response.

I appreciated your point about narrowing my scope and considering whether “trust” functions as an independent principle or as an outcome of transparency, fairness, and accountability.

In terms of scope, I have the “more precise term” you mentioned. Instead of using the broader placeholder “marginalized communities,” I was already considering focusing specifically on culturally and linguistically diverse (CLD) communities. This aligns closely with my broader line of research, where I have been exploring many topics related to CLD families, including equity in diagnostic systems as well as their educational and systemic experiences. Keeping that continuity would allow me to bring scholarly and experiential grounding to the AI ethics question in this course.

So, to be more specific, my refined research question would be:
“How can we design transparent, accountable AI systems that healthcare providers and patients from culturally and linguistically diverse (CLD) communities can trust?”

Does this seem like an appropriate and sufficiently precise narrowing in your view?

Thank you again for your guidance.

Best regards,
Piter

On Thu, Sep 4, 2025 at 8:32 AM Evan Selinger <emsgsh@rit.edu> wrote:

> Hi Piter,
>
> You’re talking to someone who very much knows what it is like to become interested in topics that are too big to tackle all at once! I’ve come to believe that this tendency is a superpower when you know how to manage it. People who only see the trees but miss the forest risk being too narrowly focused; it’s quite easy, in this mindset, to miss crucial context. The key is to create a multi-step workflow.
>
> Start with the big questions and ask yourself, “Why am I interested in them?” Write down the answers that spring to mind. Notice how I’m saying answers (plural). Once you have that list, you can shift to the next step and ask, “Will focusing on one or two items on this list narrow the scope down to something that’s both manageable and significant as a step towards addressing the big issues?” Across the disciplines, research often proceeds incrementally, step-by-step. That’s a feature, not a bug.
>
> Now, let’s think about your question, “How can we design transparent, accountable AI systems that healthcare providers and patients can trust?”
>
> The key words are “transparency,” “accountability,” and “trust.” In Unit #2, AI Ethics Essentials, you’ll see that “transparency,” “accountability,” and “trust” are three of the eleven key globally recognized AI ethics principles. So, you’re on the right track, from the start!
>
> One thing I’ve noticed in some of the discussions is that it’s debatable whether “trust” is an independent principle or what happens when the other ethical principles are met. In other words, is a trustworthy just a shorthand policy term for AI systems that are transparent, fair, accountable, and so on?
>
> Now, to bring this discussion back to your initial e-mail, it may be possible to narrow down your topic even further. You might (depending on what the literature suggests) be able to ask: “How can we design transparent, accountable AI systems that healthcare providers and patients from marginalized communities trust?” Of course, the word “marginalized” is just a placeholder here. You might find a better, more precise term.
>
> As to sources, my recommendation to use RIT’s library was merely a way to make research easy for those who haven’t conducted it before. If you have access to relevant, peer-reviewed sources from

elsewhere, that's totally fine.

Does all this make sense?

Best,
Evan

Evan Selinger
Professor, Philosophy Dept.
Rochester Institute of Technology
Office: Eastman (Building 1), Room 3146
(585) 475-2531 http://eselinger.org

From: Piter Garcia (RIT Student) <pzg8794@rit.edu>
Sent: Wednesday, September 3, 2025 9:44 PM
To: Evan Selinger <emsgsh@rit.edu>
Subject: Re: Research Topic Proposal for IDAI-700

Dear Professor Selinger,

I apologize for my delayed response; I was held back due to registration issues, which I hope will be fully resolved by tomorrow or Friday.

Thank you for acknowledging the connection between my lived experience and this research area. Since returning to school, now that I can effectively manage my condition, understanding these issues has been my primary focus. Courses like yours have guided me on the right path because each emphasizes factors that were either missing, ignored, or undervalued. With that said, let me address your questions and suggestions to ensure I understand and proceed with my research accordingly.

First, regarding your point that *"you'll need to ground any important claims in the scholarly literature"*: Yes, this is one of the main reasons I chose this course—to have time to find the necessary literature to educate myself on this topic and, significantly, to create a bridge between this research paper and my current work in *"Equitable Bioinformatics: Enhancing Diagnostic Decision-Making through RNA and Biomarker Data."* My goal is to produce work that not only amplifies my voice on this topic but also solidifies and supports it with literature and scientific data.

Second, about focusing on one of my four research questions: Thank you for pointing this out—you've identified something my brain tends to do. It often tries to cover more than it can physically accomplish because of how interesting and interconnected these factors are to something that truly matters. I will narrow it down to one question, which will likely be: ***"How can we design transparent, accountable AI systems that healthcare providers and patients can trust?"***

Regarding your suggestion to *"focus on a general issue (like cultural biases in training data, ethical frameworks for equitable outcomes, designing trustworthy diagnostic systems, or giving affected communities meaningful participation)"*: I completely agree. I have had similar discussions with instructors in different research contexts, where I suggested making the topic broader in concept—such as focusing on ADHD broadly—while expressing the desire to highlight specific diagnostics that are common, difficult to diagnose, and often intertwined with ADHD symptoms, such as a chronic illness like candida overgrowth. I understand exactly what you mean and how I should approach any specific topic that deserves attention based on the research, whether it involves demographics such as Latinos or specific diagnoses.

Third, building off existing ethics scholarship rather than running experiments: Thank you for this reminder! I understand this, but my brain needs these explicit reminders to avoid trying to build an entirely new framework from scratch. *Your comment has solidified this approach in my mind, and I am confident my brain will not go down that path*. This is precisely why I am choosing the question referenced above—*my goal is to use the literature, research papers, and my existing work to propose a framework that I can later validate.*

Yes, I will follow your advice and search for six papers using the RIT library. However, I am also planning to reference other reputable resources or resources backed by reputable institutions. **Please let me know if this approach is acceptable.**

Lastly, regarding your question, *"Does this make sense?"*: Absolutely—100% yes. Everything you commented on was not only right on point but delivered with surgical precision. I deeply appreciate that you dedicated this much time and attention to my topic and provided such valuable insights.

As instructed, my next step will be to find six relevant peer-reviewed articles and discuss them briefly with you, either via email, in person, or via Zoom, whichever is more convenient when I have gathered all relevant papers. This will allow you to provide insights about the relevance and strength of the selected literature.

Thank you so much for your time and dedication.

Best,

Piter

On Mon, Sep 1, 2025 at 2:49 PM Evan Selinger <emsgsh@rit.edu> wrote:

> Hi Piter,
>
> It's nice to hear from you!
>
> You've picked a strong research topic, "AI Ethics in Equitable Diagnostics for Marginalized Populations."
>
> I can appreciate the connections that you've drawn between your lived experience and this area of academic research. In order to write a scholarly paper that draws from sufficient scholarly research (6 peer-reviewed articles in your research portfolio), three things will be helpful.
>
> First, use your lived experience as a guide for finding appropriate peer-reviewed articles, but be careful not to treat your own experience as necessarily statistically significant. What you've encountered may very well be common. But you'll need to ground any important claims in the scholarly literature, i.e., show that the literature supports them.
>
> Second, a good paper on this topic will be well-scoped. By that, I mean you should focus on one of your four research questions. Trying to tackle more than one will likely result in an overly broad analysis. I don't want to tell you which one to choose because that's a personal decision.
>
> One thing to keep in mind is that I'm not sure how much scholarship exists on "urgent problems" involving AI that specifically affect "Latino/Hispanic" populations. There may be a substantial amount, or there may be very little. If you're specifically interested in this demographic, but there isn't much scholarly literature on it, you should focus on a general issue (like cultural biases in training data, ethical frameworks for equitable outcomes, designing trustworthy diagnostic systems, or giving affected communities the meaningful participation they deserve). And then, as is appropriate, you can bring the conversation back towards your target demographics. In other words, while the "Latino/Hispanic" populations aren't the same as marginalized groups in general, there may be some key ethical features that apply broadly.
>
> Third, please keep in mind that this is a conceptual ethics paper. Since you won't be running an experiment, you'll be building off existing ethics scholarship.
>
> Once you narrow down your research question, go to https://www.rit.edu/library/, enter the relevant search terms, and restrict the search by clicking on the "Scholarly & Peer Reviewed" box.
>
> Does this make sense? If not, I'm happy to discuss the matter further! If so, the next step is to look for 6 relevant peer-reviewed articles. You should read them carefully and see if they provide a firm foundation to build upon. You can then let me know what you've selected and your analysis of them. I'm happy to discuss whether they provide a strong foundation for your paper.

Best,
Evan

Evan Selinger
Professor, Philosophy Dept.
Rochester Institute of Technology
Office: Eastman (Building 1), Room 3146
(585) 475-2531 http://eselinger.org

From: Piter Garcia (RIT Student) <pzg8794@rit.edu>
Sent: Monday, September 1, 2025 12:46 PM
To: Evan Selinger <emsgsh@rit.edu>
Subject: Research Topic Proposal for IDAI-700

Dear Professor Selinger,

I hope this message finds you well. I am writing to propose my research topic for IDAI-700: Ethics of Artificial Intelligence, and would appreciate your feedback on its suitability for the course.

Proposed Research Topic:
"AI Ethics in Equitable Medical Diagnostics for Marginalized Populations"

Rationale for This Topic:

Personal and Academic Foundation
This topic connects directly to my lived experience as a Dominican immigrant who has navigated diagnostic delays and cultural barriers in healthcare systems. My personal journey with ADHD diagnosis, chronic illness, and neurodivergent identity in a culturally and linguistically diverse (CLD) community provides authentic insight into the real-world implications of biased diagnostic systems.

Technical Expertise
My current research background provides strong technical grounding for this work:

- **ISTE-780 Equitable Bioinformatics**: Developed the EQUITAS framework for fairness-aware algorithms
- **RNA Bias Detection Lab**: Created interactive tools for identifying bias in genetic analysis
- **GA Research Position**: Working on adversarial neural networks and multi-armed bandits
- **Advanced Programming Skills**: Proficiency in R, Python, and machine learning frameworks

Contemporary Relevance
Medical AI systems are increasingly deployed in clinical settings, yet research consistently shows these systems can perpetuate healthcare disparities. This topic addresses urgent problems affecting Latino/Hispanic populations, ADHD diagnosis in multilingual communities, and AI-driven healthcare decisions that disadvantage marginalized groups.

Research Questions I'd Like to Explore:

- How do cultural biases in training data affect diagnostic AI accuracy for marginalized populations?
- What ethical frameworks should govern AI-assisted medical diagnosis to ensure equitable outcomes?
- How can we design transparent, accountable AI systems that healthcare providers and patients can trust?
- What role should affected communities play in AI system development and validation?

Academic and Professional Alignment
This research aligns strategically with my PhD preparation goals in equitable diagnostics and inclusive AI. It builds on my existing technical skills while developing expertise in AI ethics, healthcare policy, and community-engaged research methods.

I believe this topic will allow me to produce rigorous, impactful research that advances the technical and ethical dimensions of AI in healthcare while maintaining strong connections to affected communities. The work has potential for real-world implementation that could improve diagnostic equity for marginalized populations.

I am open to any suggestions or modifications you might recommend.

Thank you for your time and consideration.

Best regards,
Piter Garcia